Tanya Djavaherpour

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Education

McMaster University Hamilton, Ontario, Canada
M.Sc., Computer Science 2023-2025
Thesis: Optimizing Genetic Programming Agents with TPG and Memory Structures

Amirkabir University of Technology (Tehran Polytechnic) Tehran, Iran
B.Sc., Computer Engineering 2018 – 2023
Thesis: Design and Implementation of a Software System for Underwater Image Processing.

Research Interests

Reinforcement Learning, Machine Learning
Genetic Programming, Evolutionary Computation, Neuroimaging Data Analysis

Research Experience

Research Assistant, McMaster University Mar. 2025 – Present
Research on neural basis of language and speech, using computational models.
Under the supervision of Dr. Christian Brodbeck at Brodbeck Lab.

Graduate Student Research Assistant, McMaster University Sep. 2023 – Apr. 2025
Research on evaluating Tangled Program Graphs (TPG) as a framework to solve Deep RL tasks.
Under the supervision of Dr. Stephen Kelly at Creative Algorithms Lab.

Research Intern, IPM Institute for Research in Fundamental Sciences Jul. 2021 – Oct. 2021
Research on time series data, testing DL architectures and developing hybrid models by integrating components of existing frameworks.

Teaching Experiences

Teaching Assistant, McMaster University

Software Design IV - Capstone Design Project (SFWRENG 4G06 A)	Fall 2025, Winter 2026
Data Structures and Algorithms for Mechatronics (MECHTRON 2MD3)	Winter 2024, 2025
Software Development (SFWRENG 3K04)	Fall 2024
Computer Architecture (CS 2GA3)	Fall 2023, 2024

Teaching Assistant, Amirkabir University of Technology

Principles of Artificial Intelligence	Fall 2022
Advanced Programming	Spring 2022
Algorithm Design	Spring 2022
Microprocessor and Assembly Language	Spring 2022
Fundamentals of Computer Programming	Fall 2020, 2021

Publications

Genetic Encoding and Shared Knowledge in Reinforcement Learning with Structured Memory 2025
T. Djavaherpour, A. Naqvi, R. Norouziani, Q. Vecher, S. Kelly. Artificial Life

Integrating Neuroplasticity into Genetic Programming Agents for Adaptive Decision Making 2025
A. Naqvi, **T. Djavaherpour**, Q. Vecher, S. Kelly. Artificial Life

MAPLE: Multi-Action Programs through Linear Evolution for Continuous Multi-Action Reinforcement Learning 2025
Q. Vacher, S. Kelly, A. Naqvi, N. Beuve, **T. Djavaherpour**, M. Dardaillon, K. Desnos. The Genetic and Evolutionary Computation Conference (GECCO)

Evolving Many-Model Agents with Vector and Matrix Operations in Tangled Program Graphs 2024

T. Djavaherpour, A. Naqvi, E. Zhuang, S. Kelly, *Genetic Programming Theory & Practice XXI.*
Tangled Program Graphs with Indexed Memory in Control Tasks with Short Time Dependencies 2024

T. Djavaherpour, A. Naqvi, S. Kelly, 16th International Conference on Evolutionary Computation Theory and Applications (ECTA).
Investigation of Sadness on Brain Mathematical Ability Using Musical and Semantical Excitation 2019

A. Davoodi Moghadam, A. Jamali, **T. Djavaherpour**, B. Taghibeyglou, 8th Conference of Basic and Clinical Neuroscience Congress, Razi Hall, Tehran, Iran.

Presentations

Genetic Encoding and Shared Knowledge in Reinforcement Learning with Structured Memory 2025
Paper Poster Presentation, Artificial Life 2025, Kyoto, Japan.

Enhancing Collaboration in Tangled Program Graphs with Shared Memory for MuJoCo Continuous Control Tasks 2025
Lightning Talk, 5 Minute Thesis (5MT), Women in Science and Engineering (WISE), University of Toronto, Toronto, Canada.

Tangled Program Graphs with Indexed Memory in Control Tasks with Short Time Dependencies 2025
Seminar Presentation, Vaader Seminars, IETR, University of Rennes, Rennes, France.

Tangled Program Graphs with Indexed Memory in Control Tasks with Short Time Dependencies 2024
Conference Presentation,16th International Conference on Evolutionary Computation Theory and Applications (ECTA), Porto, Portugal.

Tangled Program Graphs with Indexed Memory in Control Tasks with Short Time Dependencies 2024
Poster Presentation, 7th Computing and Software Poster and Demo Competition, McMaster University, Hamilton, Canada.

Notable Projects

Software Design, ANN: An image classification system using an ANN trained on the CIFAR-10 dataset, with comprehensive testing, automation, and detailed software documentation
Link to GitHub May 2024

Snail Jumper: An Evolutionary Game with Genetic Algorithm and Neural Network Link to GitHub Jun 2022

Plants vs. Zombies Game: A Single and Multiplayer Game Written in Java Using Swing and Graphics 2D Link to GitHub Feb. 2021

Technical Skills

Programming Languages: Python, C++, Java, MATLAB, SQL
Software Engineering: Software Architecture, OOP, Version Control, Familiar: Design Patterns
Tools & Frameworks: PyTorch, TensorFlow, NumPy, Pandas, Scikit-learn, MuJoCo, OpenAI Gym
Version Control: Git, GitHub/Gitlab
Complementary Agile and XP Collaboration, Team Leadership, Technical Communication

Honors and Awards

EGS Travel and Professional Development Award, McMaster University Engineering Graduate Society 2024

3rd Place Poster in the 7th Computing and Software Poster and Demo Competition, McMaster University 2024

Received a 2-Year Full Scholarship (Graduate Scholarship and Research Scholarship)
Valued at 34K CAD, McMaster University 2023